



SERANGOON JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

CHEMISTRY
JC2 Preliminary Examination
Paper 2 Structured Questions

9729/02
13 Sept 2017
2 hours

Candidates answer on the Question Paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.
The use of an approved scientific calculator is expected, where appropriate.
A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [] at the end of each question or part questions.

For Examiner's Use	
1	/ 8
2	/ 18
3	/ 25
4	/ 24
TOTAL	/ 75

This document consists of 24 printed pages

- 1** The size of an atom can be measured using the distance between the nuclei of two atoms.

For example, the 'metallic radius' of the Na atom is half the distance between two Na atoms in the crystal lattice of the metal. The 'covalent radius' of the Cl atom is taken to be half the distance between the nuclei in a Cl_2 molecule. Finally, the 'van der Waals' radius' of the Ar atom is assumed to be half the distance between two atoms in the solid state.

These three types of radius are commonly known as 'atomic radii' and the corresponding values for Period 3 elements can be found in the Data Booklet.

- (a) (i)** Explain the general trend in atomic radius across Period 3.

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[2]

- (ii)** Suggest a reason for the anomaly in the measurement of atomic radius for argon.

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[1]

- (b) (i)** State the general relationship between atomic radius and first ionisation energy across Period 3.

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[1]

(ii) Explain why the relationship does not hold from P to S.

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[1]

(c) Across period 3, the elements changes from metallic to non-metallic nature. Correspondingly, the difference in electronegativity between the elements and the oxide decreases, giving rise to different types of oxides.

A, **B** and **C** are Period 3 elements, from Na to S, inclusive.

- **A** has the highest melting point among Period 3 elements.
- **B** has the highest electrical conductivity in Period 3.
- **C** burns in air with a coloured flame.
- **B** and **C** can show the same oxidation state in their compounds.

Identify **A**, **B** and **C** and hence, write equations, including state symbols, to show the acid-base properties of their oxides.

[Give the chemical formulae for only the highest oxide formed.]

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[Total : 8]

2 Benzoic acid occurs naturally in many plants and it has many uses. It is an important precursor for the industrial synthesis of many organic substances.

(a) The benzoic acid and sodium benzoate buffer is commonly used as food preservatives in many acidic foods such as salad dressings and carbonated drinks.

(i) With the aid of relevant chemical equations, explain how the system can control pH when a small amount of base is added.

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[1]

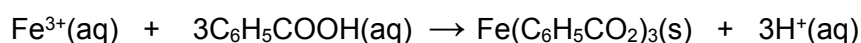
(ii) Given that the acid dissociation constant for benzoic acid is $6.3 \times 10^{-5} \text{ mol dm}^{-3}$, calculate the mass of solid sodium benzoate required to be added to 250 cm^3 of 0.50 mol dm^{-3} of benzoic acid to form a buffer solution with an initial pH 4.

[2]

- (iii) To achieve the optimal buffering capacity of a 25 cm³ sample of the buffer prepared in (a)(ii), a standard solution of 1.5 mol dm⁻³ sodium hydroxide was added. Calculate the volume of sodium hydroxide needed.

[3]

- (b) The presence of benzoic acid in plant extracts can be confirmed using neutral iron(III) chloride according to the following equation.



When 50 cm³ iron(III) chloride is added to an equal volume of the plant extract, 0.0532 g of iron(III) benzoate, Fe(C₆H₅CO₂)₃ precipitate is formed.

The resultant mixture has a pH value of 2.33.

- (i) Calculate the amount of benzoic acid that has reacted with neutral iron(III) chloride solution and H⁺ in the mixture.

[1]

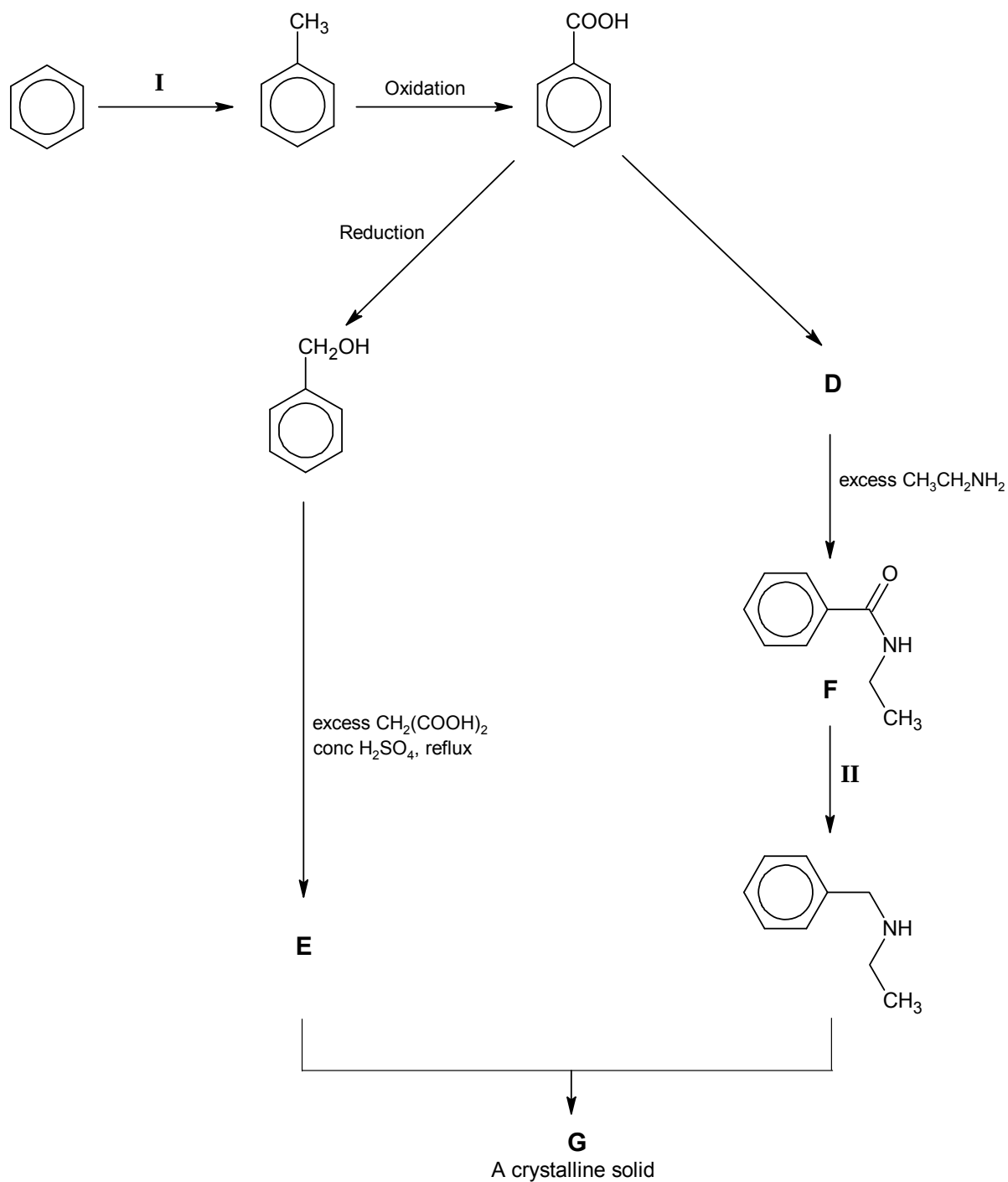
- (ii) Assuming that the H^+ ions in solution are formed from the dissociation of benzoic acid and the reaction of benzoic acid with iron(III) chloride, calculate the concentration of benzoate ions in the equilibrium mixture.

[2]

- (iii) Calculate the K_{sp} of iron(III) benzoate, given that the concentration of iron(III) ions in the mixture is $1.83 \times 10^{-3} \text{ mol dm}^{-3}$ at equilibrium.

[1]

(c) Benzoic acid is involved in the reaction scheme below:



- (i) Draw the structures for **D**, **E** and **G** in the boxes below.

D	E
G	

[3]

- (ii) Suggest the reagents and conditions for reactions **I** and **II**.

Reaction **I**:

Reaction **II**:

[2]

- (iii) Compound **F** was heated with aqueous sodium hydroxide and excess carbon dioxide was bubbled through the solution.

Given the acid dissociation constants below, suggest the products that would be formed after the introduction of carbon dioxide.

acid	formula	$K_a / \text{mol dm}^{-3}$
carbonic acid	H_2CO_3	4.5×10^{-7}
benzoic acid	$\text{C}_6\text{H}_5\text{COOH}$	6.3×10^{-5}
ethylamine salt	$\text{CH}_3\text{CH}_2\text{NH}_3^+$	1.6×10^{-11}

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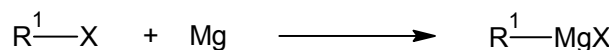
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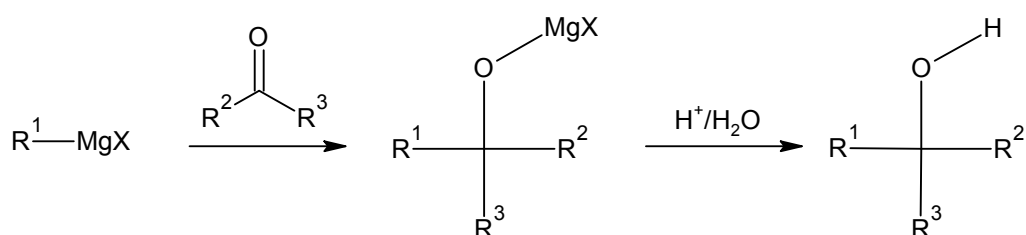
[Total : 18]

- 3 Grignard reagents are organomagnesium halides, RMgX commonly used in organic synthesis. The R in RMgX behaves like an anion, R^- and is a strong Lewis base. It can be made by the reaction between halogenoalkane and magnesium under anhydrous conditions, using ether as a solvent.



where R = alkyl group, X = Cl, Br, I

In a Grignard reaction, the Grignard reagent behaves as a nucleophile and may be added to a carbonyl compound for the formation of an alcohol. This reaction is useful for the formation of carbon–carbon bonds.



- (a) (i) Suggest, with an equation, why the synthesis of Grignard reagent has to be done under anhydrous conditions.

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[2]

- (ii) Besides organomagnesium compounds, organolithium compounds can be formed between lithium and relevant organic substances. Account for the similarity in the reactions involving lithium and magnesium.

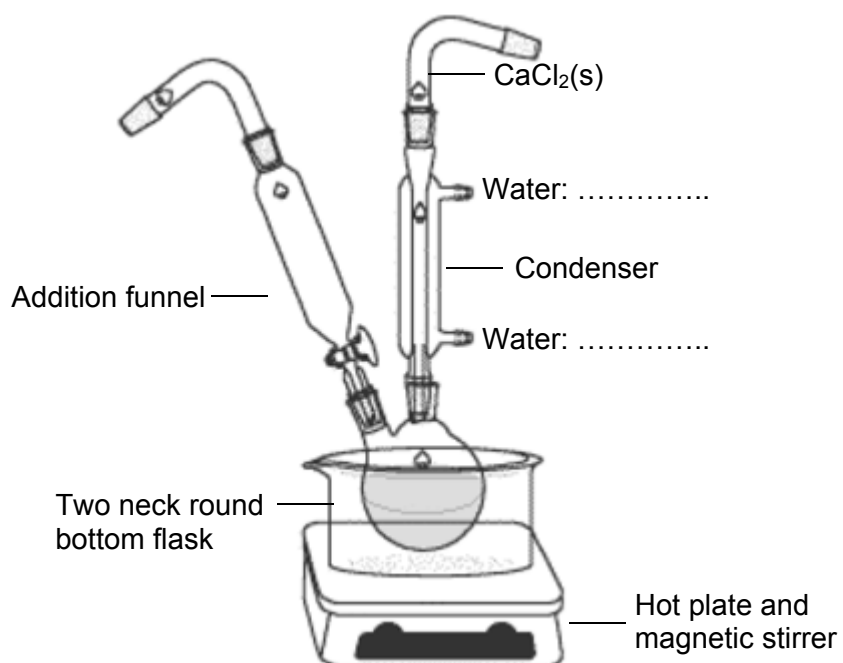
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[2]

The preparation and set up of Grignard synthesis of 2-methylhexan-2-ol is described below:

Table 3.1

Compound	Melting pt / °C	Boiling pt / °C	Molar mass / g mol ⁻¹	Density / g cm ⁻³	Solubility in water
1-bromobutane	-11.2	101.4	136.9	1.27	Insoluble
Magnesium	651	1100	24.3	1.74	Insoluble
Diethyl ether	-116.3	34.6	74.0	0.713	Insoluble
Propanone	-94.7	56.1	58.0	0.785	Soluble
Sulfuric acid	10.3	337	98.1	1.84	Soluble
2-methylhexan-2-ol	-	143	116.0	0.82	Soluble



Preparation of Grignard reagent, RMgX

1. Add 15 g of 1-bromobutane and 3 g of magnesium into the 250 cm³ two-neck round bottom flask.
2. Add 50 cm³ of anhydrous diethyl ether using the addition funnel. Gently stir the mixture under reflux for 30 minutes. The reaction is exothermic and will boil without external heat.
3. When the reaction is complete, cool the reaction mixture using an ice bath.

- (b) (i) With reference to Table 3.1, suggest why a reflux set up is necessary in this organic synthesis.

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[1]

- (ii) On the diagram on page 10, label and account for the direction of water flow.

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[1]

- (iii) Some heating may be required at the beginning as the initial rate of formation of RMgX is slow. Magnesium metal has to be scrapped or sliced into smaller pieces to increase the reaction rate.

Write an equation to suggest why magnesium may not be as reactive as expected.

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[1]

- (iv) Excess diethyl ether is added to keep the Grignard reagent at a low concentration. This will prevent the Grignard reagent from reacting with unreacted 1-bromobutane. Draw the structure of the organic by-product.

[1]

Synthesis of alcohol

4. Using the addition funnel, add 6.5 g (dissolved in 15 cm³ of diethyl ether) of propanone to the Grignard reagent.
5. To a separate 250 cm³ beaker, add 100 cm³ of ice water and 4 cm³ of concentrated sulfuric acid and mix well.
6. When the reaction with propanone is completed, add the reaction mixture into a 250 cm³ beaker, sitting in an ice bath. Carefully transfer sulfuric acid solution from step 5 into reaction mixture. Stir gently to mix well.

(c) (i) Using Table 3.1, calculate the theoretical mass of 2-methylhexan-2-ol.

[2]

(ii) The actual yield for 2-methylhexan-2-ol is 4 g. Calculate the percentage yield for this reaction.

[1]

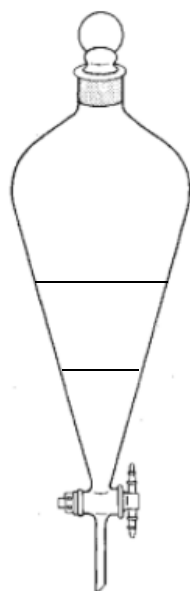
(iii) Suggest why the addition of sulfuric acid has to be carried out in an ice bath in step 6.

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[1]

Solvent extraction

7. Add the mixture from step 6 into a 500 cm³ separation funnel and add 10 cm³ of diethyl ether. Shake the separation funnel, releasing any gas formed. Allow the layers to separate and collect both layers separately.
 8. Repeat step 7 with the aqueous layer and a fresh sample of 10 cm³ diethyl ether. Collect and combine each ether layer.
 9. The combined ether layer is poured back to the separation funnel and mixed with several 20 cm³ portions of 5 % sodium bicarbonate solution until no more carbon dioxide is generated.
- (d) (i) After step 7, there are two layers in the separation funnel. Identify the ether and aqueous layer in the diagram below.



Upper layer:

Bottom layer:

Explain your answer.

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[1]

- (ii) Give a reason for the addition of sodium bicarbonate in Step 9.

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[1]

- (iii) During solvent extraction, both propanone and 2-methylhexan-2-ol are found in the ether layer. Comment on the solubility of 2-methylhexan-2-ol in diethyl ether.

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[1]

- (iv) Suggest different chemical tests to positively identify propanone and 2-methylhexan-2-ol in the ether layer.

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[2]

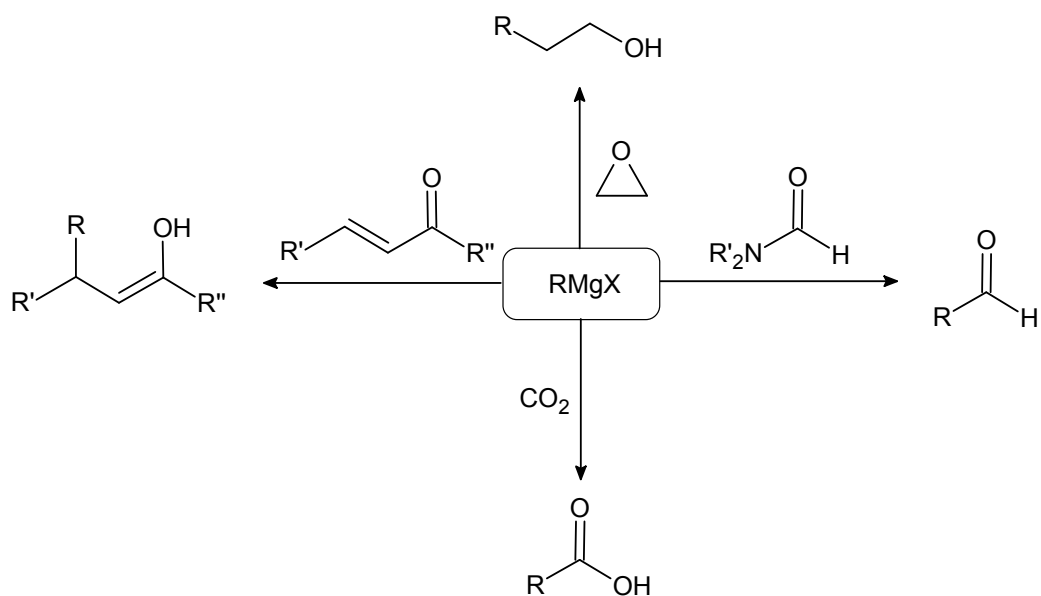
- (v) Suggest how crude 2-methylhexan-2-ol can be purified from the ether layer.

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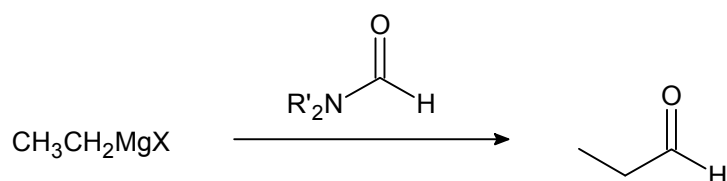
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[1]

- (e) Grignard reagent is used widely in organic synthesis due to its ability to increase carbon chain length. Four unique reactions involving the use of Grignard reagent are shown in the reaction map below.



- (i) The following Grignard reaction to form propanal achieved a very low yield as the aldehyde formed can react with the Grignard reagent again.



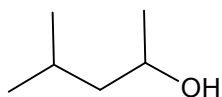
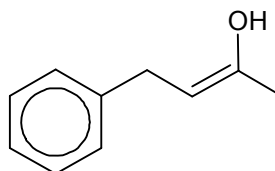
Describe the mechanism between propanal and the Grignard reagent, $\text{CH}_3\text{CH}_2\text{MgX}$ to form the corresponding alcohol.

[2]

- (ii) Starting with a suitable Grignard reagent, propose an alternative 3-step synthesis pathway to synthesise propanal.

[3]

- (iii) Using different reaction types from the reaction map on page 15, suggest appropriate pairs of reagents in the synthesis of compounds **H** and **I**.

**H****I**

Reagent pair for H:

Reagent pair for I:

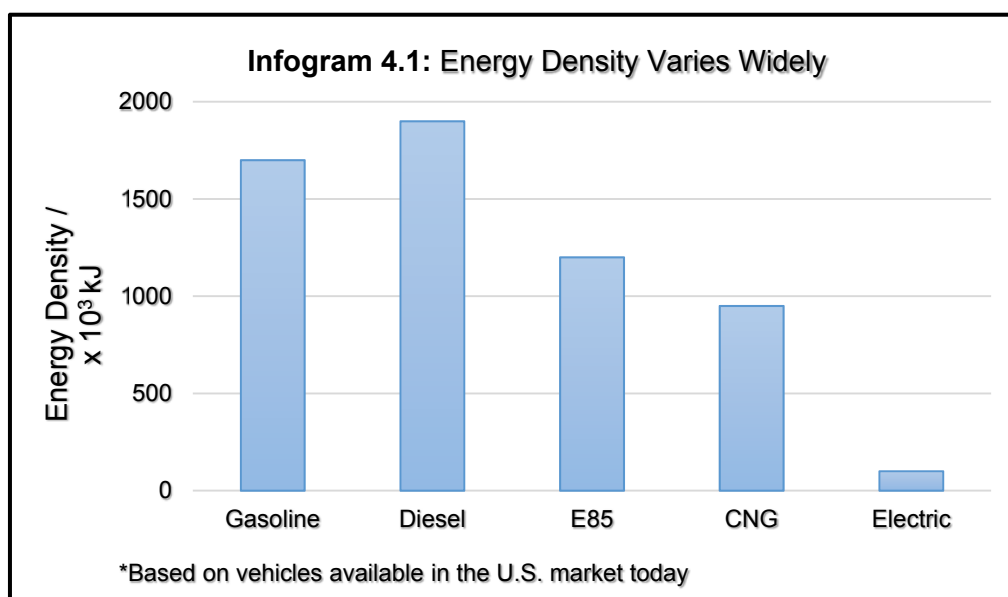
[2]

[Total: 25]

- 4 The use of electric vehicles has been increasing popular in the U.S. For example, hydrogen fuel cell vehicles produces zero carbon emission.

Despite the advantages of electric cars, petrol-engined cars are still widely used today. One problem with electric vehicles is their poor performance, inconvenience of finding a charging station and it takes a long time to recharge the batteries. Petrol-engined cars are better as their energy supply can be replaced over 6 million times faster than electric cars.

The infogram below shows how energy density varies widely with different fuels. Energy density is the energy produced per unit dimension of a fuel and this determines the practicality of a fuel.



However, using petrol (which contains a mixture of alkanes) has its own problems. One such problem is that it's a finite resource, and another is pollution. Hence, there is a need to improve the performance of petrol in car engines so that it burns as cleanly and efficiently as possible and to find alternative sources of fuel to replace petrol.

Some data for five fuels are given in **Table 4.2**.

Fuel	Formula	Standard enthalpy change of combustion, $\Delta H_c^\ominus$ / kJ mol^{-1}	Relative molecular mass	Energy density (energy transferred on burning 1 kg of fuel) / kJ kg^{-1}
octane	$\text{C}_8\text{H}_{18} (\text{l})$	– 5470	114.0	
1,2-xylene	$\text{C}_8\text{H}_{10} (\text{l})$	– 4540	106.0	
carbon	$\text{C}(\text{s})$	– 393	12.0	
methanol	$\text{CH}_3\text{OH}(\text{l})$	– 726	32.0	
hydrogen	$\text{H}_2(\text{g})$	– 286	2.0	

- (a) Give an example of a pollutant that can arise from the use of petrol and state a detrimental effect of the pollutant.

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[1]

- (b) Complete Table 4.2 by calculating the energy density for all fuels. [1]

- (c) (i) On the basis of energy density, which is the best fuel in Table 4.2.

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[1]

- (ii) Suggest a practical difficulty in using hydrogen rather than methanol as a fuel.

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[1]

- (d) Using the given data above, comment on the feasibility of using electric cars in the near future.

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[1]

Octane is commonly used as a fuel or a precursor in the production of other chemicals. The term “octane rating” is often used in the petroleum industry for rating the ability of octane’s various branched isomers in reducing engine knock in vehicles.

One important factor in making petrol is getting the right octane rating to reduce the problem of ‘knocking’. Knocking occurs when the combustion of fuel is not smooth. Knocking reduces engine power, leading to wear and tear in the engine and results in wastage of petrol. Hence, highly branched alkanes are desirable in petrols as they have higher octane ratings and reduce knocking.

Another factor for getting the right properties of petrol is the blending of petrol. In cold weather, it is difficult to vaporise petrol, which makes ignition difficult. On the other hand, in hot weather, petrol vaporises easily. Hence, petrol companies make different blends of petrol for different seasons and climates.

You are to use the information provided above to answer parts **(e)** and **(f)**.

- (e)** Apart from using alkanes with high octane ratings, another way to deal with the problem of knocking is through the addition of an antiknock agent, tetra-ethyl lead (**TEL**). **TEL** is a highly volatile colourless liquid which aids in the formation of free radicals. However, this method is not commonly used nowadays.

Explain, with the aid of a suitable equation, how **TEL** helps to overcome the problem of knocking and give a reason why its usage is limited nowadays.

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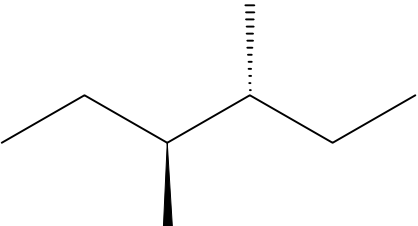
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[3]

- (f) Compounds **J**, **K** and **L** are isomers of the hydrocarbon octane, C_8H_{18} .

The table below shows the boiling points and data relating to the optical activity of the compounds.

Structure	Boiling Point / °C	Number of chiral centres	Optical Activity
Isomer J	120	1	Yes
 Isomer K	118	2	No
Isomer L	107	0	No

- (i) Suggest why isomer **K** does not exhibit enantiomerism.

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[1]

- (ii) Propose the structures of compounds **J** and **L**.

Isomer J :	Isomer L :
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[2]

- (iii) In terms of structure and bonding, identify which of your proposed compounds (**J** or **L**) would be a more suitable fuel for use in summer and winter respectively.

Your answer should show clear and detailed comparison in your choice of fuel.

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[2]

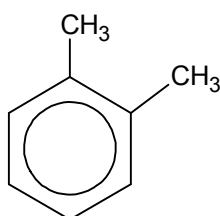
- (iv) Controlled chlorination of isomer **K** in the presence of UV light produces different mono-chlorinated products with a molecular formula of $C_8H_{17}Cl$.

Draw the structural formula of any **two** possible isomers and state the ratio in which they are formed.

[2]

- (g) The performance of hydrocarbons as fuels is largely improved by catalytic reforming. Reforming takes straight chain hydrocarbons in the C_6 to C_8 range from the gasoline or naphtha fractions and rearranges them into compounds containing benzene rings. They are used to make fuels with high octane number. Hydrogen is produced as a by-product of the reactions. A catalytic mixture of Pt and Al_2O_3 at a temperature of $500\text{ }^\circ\text{C}$ and pressure of 20 atm is used.

An example of a product is 1,2-xylene.



- (i) Write a balanced equation when octane undergoes the 'reforming' process to form 1,2-xylene.

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[1]

- (ii) What *type of reaction* has the octane undergone?

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[1]

- (iii) 1,4-xylene can be produced by a similar process. Suggest the structural formula of a C_8 alkane which might produce 1,4-xylene.

[1]

- (iv) By means of an energy cycle, show and calculate the standard enthalpy change of reaction that produces 1,2-xylene from octane.

[2]

- (v) Using data from the Data Booklet, calculate the enthalpy change of the reaction which reforms octane into 1,2-xylene.

[2]

- (vi) By considering your answers in (g)(iv) and (v), suggest if they are accurate descriptions of the reformation process.

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[1]

- (vii) Briefly explain whether you would expect the above reaction to have its ΔG value to be more negative than its corresponding ΔH_r .

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[1]

[Total : 24]